

CONTACTLESS DOORBELL USING ARDUINO

PRESENTED By :

Guided By :
MRS.LALITHA K

PRASAD R S {22AI037}
NAGAJOTHI T R {22IT024}
GOVARTHINI M {22IT009}

Abstract



1. A contactless doorbell system using Arduino and an ultrasonic sensor.
2. Detects the presence of a person near the door without physical touch.
3. When a person approaches, the ultrasonic sensor signals the Arduino.
4. Arduino triggers a buzzer or relay-controlled doorbell to sound.
5. Provides a hygienic solution, reducing the risk of germ transmission.
6. Can be integrated with smart home systems for notifications or camera activation.

Objectives:

- 1.Design a contactless doorbell system using Arduino and an ultrasonic sensor.
2. Detect a person's presence and trigger a sound alert without touch.
3. Improve hygiene by eliminating physical contact with the doorbell.
- 4.Create an efficient and scalable IoT solution for home automation.
- 5.Enable integration with smart home systems for added functionality.

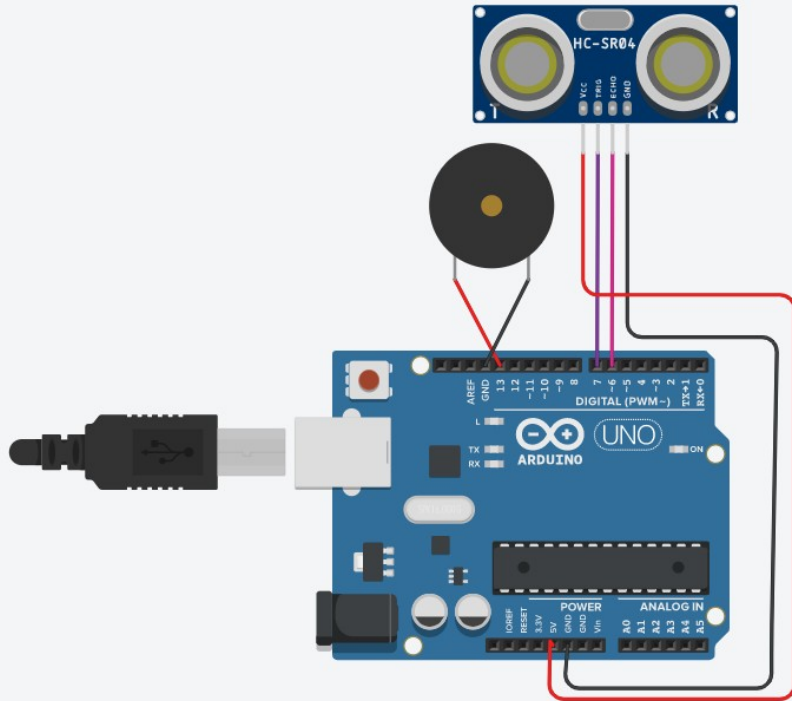


Components Required:



1. Arduino board (Uno, Nano, etc.)
2. Ultrasonic sensor (HC-SR04)
3. Buzzer or Speaker
4. Relay module (optional)
5. LEDs (optional)
6. Jumper wires
7. Breadboard
8. Power supply or USB cable

Flow Diagram:



Implementation:

(Hardware or software)

1. **Connect the Ultrasonic Sensor** : to the Arduino using jumper wires, ensuring proper connections for VCC, GND, Trig, and Echo pins.
2. **Wire the Buzzer** : or relay module to the Arduino to act as the doorbell alert mechanism.
3. **Upload the Arduino Code** : that reads distance data from the ultrasonic sensor and triggers the buzzer or relay when a person is detected within the set range.
4. **Test the System**: to ensure the sensor accurately detects proximity and the buzzer activates as expected.
5. **Optional** : Integrate LEDs for visual feedback or additional smart home features, such as notifications or camera activation.
6. **Assemble and Mount** : the components in a suitable enclosure and install the system near the door for operation.



Thank you